NUS-ISS

MASTER OF TECHNOLOGY IN ARTIFICIAL INTELLIGENCE SYSTEMS

Graduate Certificate Examination

Subject: *Pattern Recognition Systems*

Sample Examination Questions

SECTION A

Question 1 *(Total: 20 Marks)*

During the COVID pandemic, more and more consumers have begun shopping online. For ecommerce platforms, one of the most important things they want to know is whether the customers are just browsing or actually buying.

You have recently been employed as an AI engineer by an online bookstore. You have started a project on predicting the purchasing intention of visitors using aggregated pageview data gathered during the visit along with some session and user information. If a subset of users are predicted as more likely to purchase, the bookstore can take proactive actions such as offering time-limited coupons to push the users to complete their purchases. This will help the bookstore target the right customers, improve retention and ultimately increase sales and profits.

In this project, the purchasing intention model is designed as a binary classification problem measuring the user's intention to finalize the transaction. A historical dataset has been collected in a one year period, which consists of feature vectors belonging to 12,330 sessions. Of these sessions, 84.5% (10,422) were negative class samples that did not end with purchasing, and the rest (1,908) were positive class samples that ended with purchasing.

The numerical and categorical features of the dataset are shown in Tables 1 and 2, respectively. The "Bounce Rate", "Exit Rate" and "Page Value" features represent the metrics measured by Google Analytics for each page in the website. The value of "Bounce Rate" feature for a web page refers to the percentage of visitors who enter the site from that page and then leave ("bounce") the site without triggering any other requests including shipping and payment to the analytics server during that session. The value of "Exit Rate" feature for a specific web page is calculated as for all pageviews to the page, the percentage that were the last in the session. The "Page Value" feature represents the average value for a web page that that a user visited before completing a transaction. It measures the page's economic value to the bookstore.

Table 1 Numerical features of the dataset

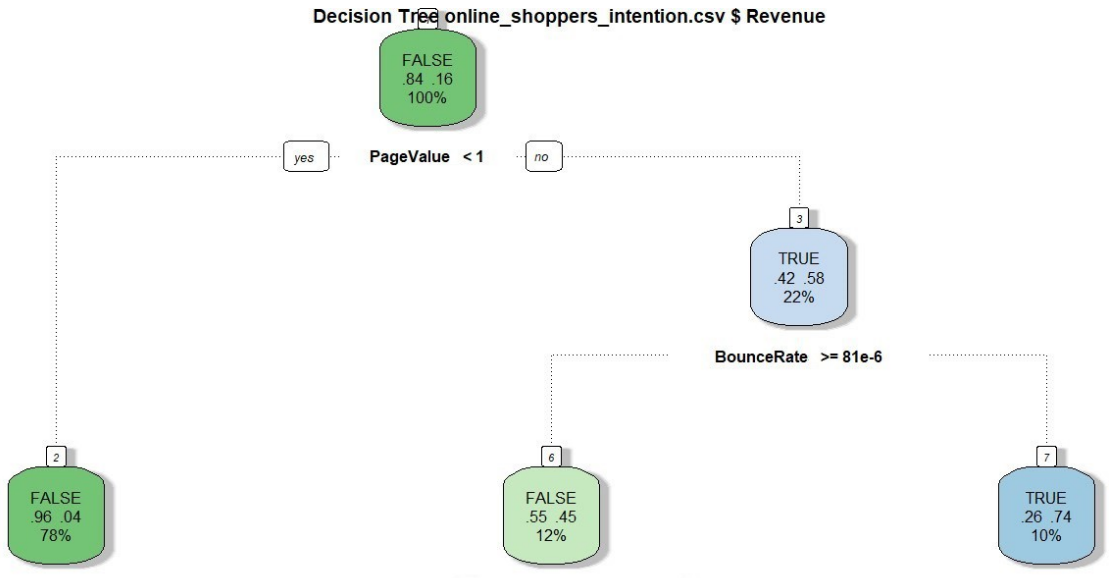
Feature name	Feature description	Min Value	Max Value	Mean	Standard Deviation
Administrative	Number of pages visited by the visitor about account management	0	27	2.32	3.32
Administrative duration	Total amount of time (in seconds) spent by the visitor on account management relatedpages	0	3398	80.82	176.70
Informational	Number of pages visited by the visitor about Web site, communication and addressinformation of the shopping site	0	24	0.50	1.26
Informational duration	Total amount of time (in seconds) spent by the visitor on informational pages	0	2549	34.47	140.64
Product related	Number of pages visited by visitor about product related pages	0	705	31.73	44.45
Product related duration	Total amount of time (in seconds) spent by the visitor on product related pages	0	63,973	1194.75	1912.25
Bounce rate	Average bounce rate value of the pages visited by the visitor	0	0.2	0.02	0.04
Exit rate	Average exit rate value of the pages visited by the visitor	0	0.2	0.04	0.05
Page value	Average page value of the pages visited by the visitor	0	361	5.89	18.55
Special day	Closeness of the site visiting time to a special day	0	1.0	0.06	0.19

Table 2 Categorical features of the dataset

Feature name	Feature description	Number of categorical values
OperatingSystems	Operating system of the visitor	8
Browser	Browser of the visitor	13
Region	Geographic region from which the session has been started by the visitor	9
TrafficType	Traffic source by which the visitor has arrived at the Web site (e.g., banner, SMS, direct)	20
VisitorType	Visitor type as ‘‘New Visitor,’’ ‘‘Returning Visitor,’’ and ‘‘Other’’	3
Weekend	Boolean value indicating whether the date of the visit is weekend	2
Month	Month value of the visit date	12
Revenue	Class label indicating whether the visit has been finalized with a transaction	2

Answer the following questions:

a. You start to build decision tree models to predict user purchase intention. The Revenue feature is used as the class label which can be only TRUE or FALSE. TRUE means the user purchased and finalized with transaction. The rest of the variables are used as input variables. 70% of the historical data is used to train the model and the model obtained is shown below. What does this model tell you about the common characteristics of the users who are more likely to purchase? How does this analysis relate to the business outcomes described in the case study? Identify one possible problem with the current model and suggest how it might be improved. How would you determine if your improved model is more valuable to the bookstore?



(4 Marks)

b. You continue to build Support Vector Machine (SVM) and Neural Network (NN) models for the same task in part (a). You have used a linear kernel for SVM and one hidden layer with 10 hidden nodes for NN. The tables below show the two confusion matrixes after testing these two models on the test data, which is 30% of the historical data. Compare the performance of these two models in the context of bookstore business objectives. State which model best meets the business objectives and give your reasons.

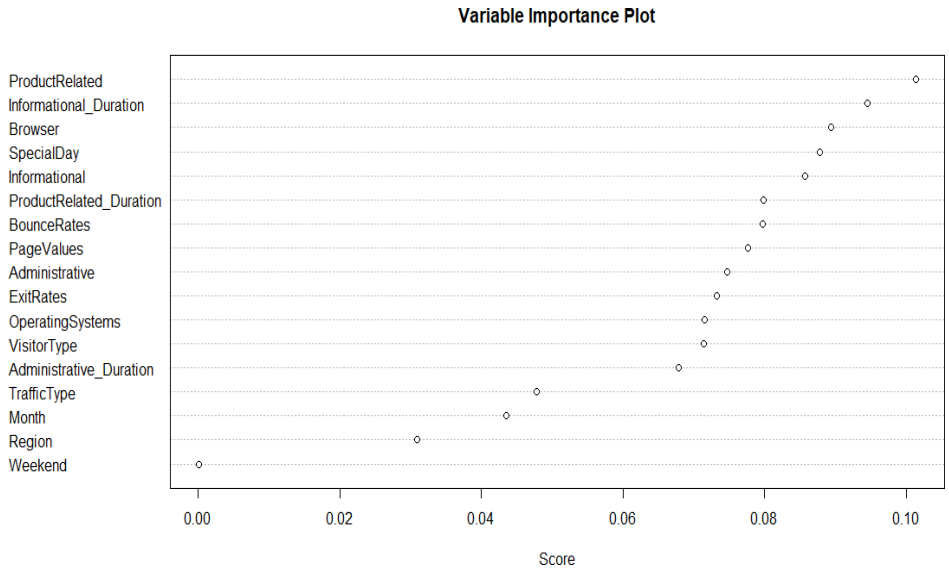
SVM Model	Predicted True	Predicted False
Actual True	3078	72
Actual False	312	237

NN Model	Predicted True	Predicted False
Actual True	3029	121
Actual False	261	288

(3 Marks)

c. You have also built an AdaBoost model using decision tree as base model and the variable importance is obtained from the model as shown below -- The higher the score, the more important the variable is. Compare the variable importance from the AdaBoost model with that you may derive from the model in part (a). You need to highlight the difference and provide possible reasons for such difference. Propose a possible way to make use of such importance information for further modelling.

(3 Marks)



d. You have been further provided with detailed sequential clickstream data, which includes the pages users visited, time spent on each page, how users arrived on the page, and where they went next. It is suggested that you can use the clickstream data to build sequential models to predict the probability of visitor's intention to purchase. Propose a suitable sequential modelling technique for this purpose. Briefly describe how such a model can be built, including the training data structure and pre-processing required. How can it be used together with the models built earlier to achieve better prediction results?

(4 Marks)

e. The company's Chief AI Scientist postulates that even though there are 18 features, there could only be a handful of underlying factors since some of the features are intuitively related to each other. He has asked you to investigate if this is true. You decided to perform a PCA which resulted in the following output.

components	Informational_Duration	Informational	ProductRelated	ProductRelated_Duration	BounceRates	ExitRates	PageValues	
PC1	0.31656	0.36009	0.44827	0.44868	-0.35206	-0.39517	0.15834	
PC2	0.35695	0.33731	0.23813	0.27445	0.47291	0.45402	-0.31092	
PC3	0.04342	0.03120	0.04760	0.06076	0.17954	0.14909	0.30342	
PC4	0.06585	0.05851	-0.06828	-0.05430	0.32626	0.27815	0.53560	
PC5	0.42894	0.37502	-0.37942	-0.34231	-0.05050	-0.03814	-0.02708	
PC6	-0.02662	-0.01862	0.00397	-0.01649	-0.03671	-0.03640	0.03506	
PC7	-0.34011	-0.27146	0.30944	0.28889	0.16838	0.12140	-0.07580	
PC8	0.05355	0.04139	-0.00877	-0.01067	-0.00171	0.00491	0.09538	
PC9	0.03410	0.00984	-0.01949	-0.02177	0.04435	0.05456	0.02439	
PC10	0.01737	-0.01236	0.01420	0.02659	-0.00885	0.00862	-0.06897	

components	SpecialDay	OperatingSystems	Browser	Region	TrafficType	Weekend	Revenue	%var_expl
PC1	-0.08056	-0.01912	-0.01350	-0.02678	-0.06575	0.04364	0.22069	20.7
PC2	0.11484	-0.00746	-0.08074	-0.07482	0.02993	-0.04733	-0.26803	12.3
PC3	0.02441	0.52408	0.48995	0.23968	0.44804	-0.05745	0.26050	10
PC4	-0.12396	-0.29912	-0.25082	-0.18921	-0.15836	0.01415	0.53397	9.3
PC5	-0.37612	0.05141	-0.00070	0.25960	-0.00512	0.43780	-0.07941	7.5
PC6	0.50667	0.08746	-0.17268	-0.36361	0.32323	0.67961	0.04490	7.2
PC7	-0.40535	0.00071	-0.05112	0.34346	-0.11813	0.53265	0.00538	6.9
PC8	0.59718	-0.24090	-0.09143	0.73231	-0.14907	0.01925	0.06498	6.7
PC9	0.20218	0.12956	0.57927	-0.20887	-0.71436	0.21237	0.01013	6.2
PC10	-0.05619	-0.74032	0.55954	-0.05067	0.34339	0.09399	-0.00971	5.4

(i) How many underlying factors can you detect? Provide justification for your choice.

(3 Marks)

(ii) Explain what each factor stands for, and discuss whether these factors can be used as predictor variables in a downstream regression model.

(3 Marks)

Question 2 *(Total: 20 Marks)*

The global generation of waste electrical and electronic equipment (WEEE), or e-waste, is estimated to be 30–50 million tons per year. Although WEEE has a high recycling value, many WEEE items also contain hazardous substances that must be separated from the municipal waste stream. Waste collection companies rely on the involvement of society to properly dispose of e-waste. Different collection methods have been offered, which include collection at supermarkets and electrical and electronic equipment stores, and municipal collection centers. Other methods include mobile collection, such as curbside collection and on-demand collection, where resident can request that the material be collected from a household. On-demand collection may be suitable for efficiently planned waste collection, especially in city centers.

In the collection of e-waste, sorting e-waste into categories for downstream recycling processes is labour-intensive. Machinew, an AI company founded in Singapore, is considering building a robotic sorting system for small appliances. The system will use an artificial intelligence to identify objects on a conveyor belt. The AI will identify the brand of the appliances, the type of the electrical appliances, and the recycling value of the electrical appliances (See Table 3 for more detail).

Table 3.

Brand	Type	Value
Cornell	Kettle	Low
Electrolux	Vacuum cleaner	Mid
Eurospace	Microwave oven	High
Hitachi	Rice cooker	
LG	Air Fryer	
Mayer	Food steamer	
Midea	Toaster	
Panasonic	Pressure cooker	
Samsung	Coffee machines	
Sharp	Irons	
Philips	Sewing machines	
Tefal	Sandwich maker	
Mistral	Blender	
Rowenta	Juicer	
Braun	Mixer	
Iona	Slow cooker	
Sona	Hair dryer	
Bosch		

After the identification, a series of robotics arms will pick the object and place it into the correct bin. There will be a separate bin for each brand-type-value combination. The company believes the system is much more efficient compared to human workers. In general, manual picking usually involves 15 picks per minute, however, this sorting robot should be able to perform up to 70 picks every minute. This will significantly reduce operating costs and improve overall productivity.

In this exciting venture, you are assigned to a team that builds the AI system that is able to (1) identify the brand, the type and the value of the appliances on a conveyor belt using the camera mounted on the robotic arm; (2) control the robotic arm to pick the e-waste objects; (3) provide predictive maintenance for the conveyor to perform anomaly detection and anomaly diagnosis.

The entire team is reported to an AI product owner, who has requested the team to first build a lean deep learning model. So far the team has managed to collect 5,000 images, but the spread is uneven: some categories have much fewer images. The images come in various resolutions, but none of them are smaller than 256 x 256. You were told more images would come, and based on the collected images, one of your teammates has built the below model shown in Table 4.

Table 4.

Layer (type)	Output Shape	Param #
input_2 (InputLayer)	[(None, 128, 128, 3)]	0
conv2d_3 (Conv2D)	(None, 126, 126, 16)	448
max_pooling2d_3 (MaxPooling2	(None, 63, 63, 16)	0
conv2d_4 (Conv2D)	(None, 61, 61, 32)	4640
max_pooling2d_4 (MaxPooling2	(None, 30, 30, 32)	0
conv2d_5 (Conv2D)	(None, 28, 28, 48)	13872
max_pooling2d_5 (MaxPooling2	(None, 14, 14, 48)	0
flatten_1 (Flatten)	(None, 9408)	0
dense_3 (Dense)	(None, 256)	2408704
dense_4 (Dense)	(None, 128)	32896
dense_5 (Dense)	(None, 918)	118422

Total params: 2,578,982
Trainable params: 2,578,982

a. After looking at the model structure and the training outcome, the AI product owner was not satisfied with the proposed model. She found the last few layers of the model

are not well designed. (i) Do you agree with the assessment? State your reason(s). (ii) Propose a better model to address her concern. (iii) Explain the reason(s) and the advantage(s) behind the changes. (iv) Specify the change in the number of parameters with your new model, the activation function(s) you will use for the classifier part of the model and the loss function(s). Use table and/or figure to illustrate your idea if necessary.

(8 marks)

b. Machinew wants their robotic system to pick and place the target objects in an intelligent manner. Given that RGB-Depth cameras (attached to the robotic arm) are used to identify the objects on the conveyor belt and that its identification algorithm will be dependent on the classifier model as develop in part (a), you are tasked by Machinew to propose a suitable solution for their robotic system by considering the following criteria:

- The system's autonomous level is high (i.e. the robotic arms are able to make decision on their actions independently, with the help of the sensors and algorithms)
- The RGB-Depth cameras (attached to the robotic arm) are also used to detect the position and trajectory of the target objects, apart from just identifying them
- The robot is in the same movement envelope as the other robot arms
- It is safe and able to avoid collision with other objects (e.g. non-target objects, other robots and/or any humans onsite). You may use externally installed RGB-Depth cameras to detect the position and trajectory of these objects.
- The arm cannot be configured into odd positions that will risk damaging it (e.g. damaging the gripper when a bad grip point was chosen by the model, damaging the motor wires when certain set of joint angles are chosen).
- It is able to effectively and efficiently pick and place the target objects (i.e. pick from the conveyor belt and place into the correct bin) even when they are in odd positions/orientations
- It utilizes multi-layer neural network(s) for its decision making process

You decide to use deep reinforcement learning for this application. You are to:

(i) Identify 3 factors that are needed to be considered for your reward function based on the above criteria,

(ii) State how this could be undertaken by drafting an architecture to represent your solution. Note that your architecture has to include the following components: the physical robot hardware, the target object, other objects to avoid collision with, the identification system, the decision-making system and other relevant systems.

(iii) One issue that might arise during learning is using the real physical robot to train the model, which can be dangerous, considering that it would need real physical collisions to occur in order to properly train the robot to avoid them. Suggest with explanations what you can do to minimize the implications of this issue.

(4 marks)

c. The assembled AI sorting software/hardware system has mechanical components, which are subject to wear and tear and possible anomaly. To perform monitoring and maintenance, a set of vibration sensors are fixed to the conveyor belt. The **anomaly detection** mechanism notices occurrence of equipment anomaly by detecting that the equipment is different from usual. You have collected a vibration signal dataset in the past one month. The dataset has 3,000 normal vibration signal sequences and 2 abnormal vibration sequences (that rarely happen in the real operation). Each sequence has a duration of 4 seconds and a sampling rate 50 Hz. Figure 1 demonstrates the comparison between a normal and an abnormal signal sequence, including their original signal sequences, their spatial-domain histograms (10 bins), and their frequency-domain statistics (4 statistics including mean, median variance, standard deviation for each of 5 subbands of wavelet transformation, totally, 20 dimensions). To identify the abnormal sequence, your colleague has suggested you a classification approach as described in Table 5. Do you agree with this proposal? List two challenges that you could face using this proposal and suggest solutions to address these challenges. Provide your justifications in the context of data presented in Figure 1.

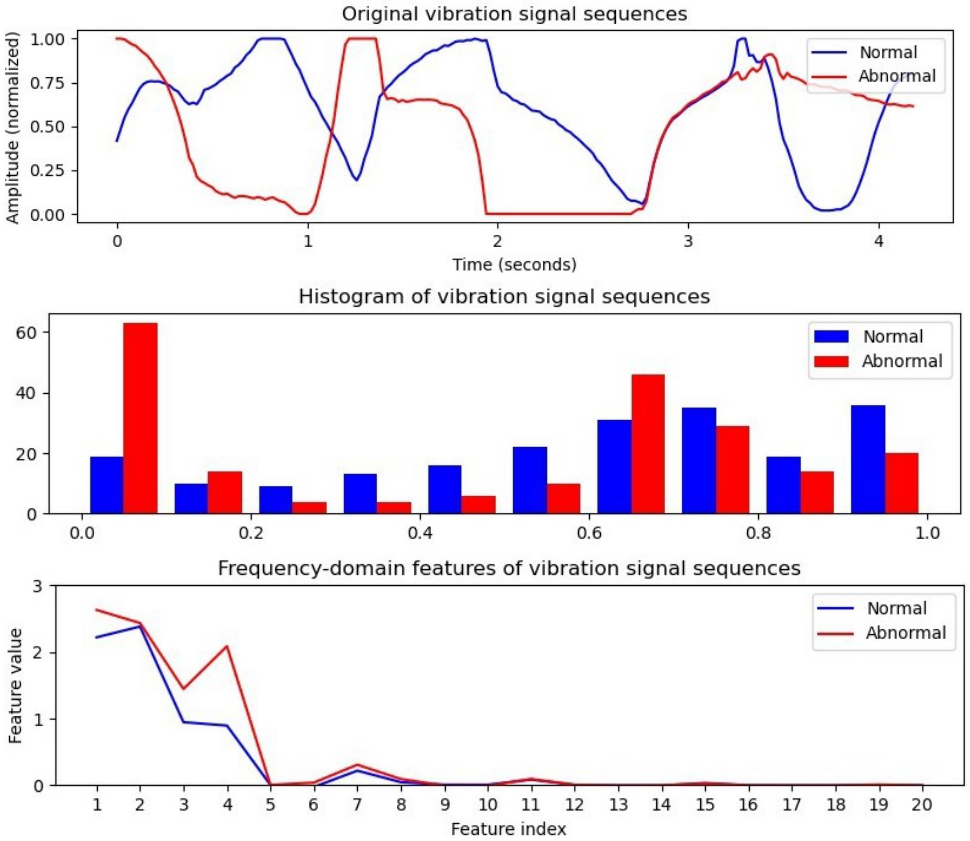


Figure 1. A comparison of normal and abnormal vibration signal sequences.

Table 5. An abnormal vibration signal detection proposal suggested by your colleague.

Feature extraction module	Firstly, extract 10-bin spatial-domain feature and 20-dimensional frequency-domain feature from the input vibration signal sequence. Then, concatenate these two features to be 30 dimensional features, which be used as input to the following classification module.
Classification module	Develop a multiple-perception-layer neural network model to perform two-category classification using your collected dataset described in the question.

(4 marks)

d. Another **anomaly diagnosis** mechanism of your developed AI system identifies the cause of equipment anomaly based on the equipment condition at that time. When an anomaly occurs, the cause is quickly isolated and recovery action is taken. Table 6 lists the maintenance records of the conveyor belt in the past one year. The vibration sensory data has three categorical measurements: 'Large', 'Fair', 'Small'. The acoustic data has two categorical measurements: 'High' and 'Low'. The failure cause is also recorded, where there are two causes 'Carryback' (some waste material remains on the belt after the electronic waste has been removed by the robot) and 'Mistracking' (the conveyor belt rides unevenly on rollers, favoring one side over the other). You are asked to build a statistical classification approach using Bayesian method based on the historical maintenance record presented in Table 6. After that, apply your approach to make decision based on the vibration measurement 'Fair' and acoustic measurement 'Low'. Show your calculation to justify your suggested approach.

Table 6. The maintenance records of the conveyor belt.

Maintenance record No.	Vibration sensory data measurement	Acoustic data measurement	Cause
1	Large	High	Carryback
2	Fair	High	Mistracking
3	Small	High	Mistracking
4	Large	High	Carryback
5	Fair	Low	Mistracking
6	Small	Low	Carryback
7	Large	Low	Mistracking
8	Fair	Low	Carryback
9	Small	Low	Carryback

(4 marks)